

TraceSurgery

Replay-controlled counterfactual interventions for failure causality and recoverability in computer-use agents.

PROBLEM

Post-hoc agent diagnoses can sound causal without proving that changing the suspected step or state would change the executed outcome.

CORE IDEA

Checkpoint a failed trajectory, replay the unmodified suffix to establish fidelity, then intervene on action, state, or continuation and re-grade the resulting world state.

VERIFIED EVIDENCE TODAY

48/48

tests

92.75%

coverage

500/500

oracle-valid tasks

Local deterministic demo: 2/2 replay controls faithful; 3 interventions change the failed outcome. This is software/causal-semantics evidence, not frontier-model performance.

INSIDE THE RESEARCH PROCESS

9 documented investigations · 3 failed repair attempts · 8 design decisions · 4 unexpected findings · 9/9 public local claims mapped to raw evidence.

MECHANISTIC FINDING

Before state corruption, action patch / suffix replan can recover while state reset alone does not survive the next bad action. After corruption, only state repair recovers the task.

EVIDENCE LAYER

Experiment journal · failed experiments · decision log · real eval tables · raw branches · qualitative case · Git history boundary · machine-readable Claim Ledger + verifier.

RESEARCH BOUNDARY

No SOTA or frontier-model performance claim is made. The remaining milestone is a matched multi-model study in realistic browser/desktop environments with repeated trials and human calibration.